ARK-444 — RESULTS

Decision-to-Execution Integrity on IBM Quantum (Heron r2)
Remnant Fieldworks Inc. — Derek Hone
Governing principle: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

VERDICT: PASS

Central question — answered: Can the system detect when an approved action is altered before execution and fail closed? On this hardware and binding, **yes.** All five preregistered PASS conditions (Field 20) were met with the in-situ SPAM baseline within ceiling on both qubits and no calibration drift. Every post-approval alteration (destination, amount, operation type, appended action) and the replayed stale approval **failed closed** — the payload did not execute — while a mutated action that was **re-verified** correctly executed.

Execution record

Item	Value
Backend	**ibm_marrakesh** (Heron r2, 156 qubits)
Instance	open-instance
Selected qubit pair (Field 10)	**Q_A = 5** (RE 0.4883%), **Q_P = 6** (RE 0.4028%), sum 0.8911%
Qualifying connected pairs	92 (min-sum rule; next best [140,141]=0.9033%)
Shots	8,192 / arm × 8 arms (principal); 2,048 × 4 (SPAM)
Transpiler	optimization_level=1, initial_layout=[5,6], **no dynamical decoupling**
Integrity gate	single-register `if_test` (`ci == 0b11`) + top-level mid-circuit `reset`
SPAM job id	`d9cmadkinv1c73anvn90`
Principal job id	`d9cmgi9htsac739bv2mg`
Calibration snapshot	`calibration_snapshot_marrakesh_20260716.json`

Ordering honored (Field 27): code+MANIFEST committed (lock `633f7fb`) → qubit pair frozen (`43a6651`) → SPAM committed (`f7dd557`) → **v1.1 control-flow re-lock** (`144fb17`) → corrected principal job id committed **before** reading (`33cd558`) → raw counts committed (`acd17f4`) → analysis (this record).

Provenance (Amendment v1.1): the v1.0 principal job `d9cmdvsjeosc73fgfk5g` (nested `if_test`) **ERRORED** with IBM code 1524 (nested conditionals unsupported) and produced **zero counts**. The integrity gate was flattened to a single-register condition (measurable semantics unchanged), re-locked at `144fb17` **before** the corrected job was submitted. This is a pre-data technical correction, not a rescue-after-failure — there was no result to rescue. The errored job id is retained permanently.

Preregistered metrics (Field 17) with 95% Wilson CIs

Metric	Definition	Value	95% CI
S_match	approved-unchanged executes (Arm 1, reference)	0.9808	[0.9776, 0.9836]
L_dest	destination-changed leakage (Arm 2, PRIMARY)	0.0087	[0.0069, 0.0109]
L_amount	amount/parameter-changed leakage (Arm 3, PRIMARY)	0.0101	[0.0082, 0.0125]
L_optype	operation-changed leakage (Arm 4, PRIMARY)	0.0082	[0.0064, 0.0104]
L_append	extra-action-appended leakage (Arm 5, PRIMARY)	0.0112	[0.0092, 0.0138]
L_replay	approval-replayed leakage (Arm 6, PRIMARY)	0.0002	[0.0001, 0.0009]
S_reverified	mutated-then-reverified executes (Arm 7, recovery)	0.9781	[0.9748, 0.9811]
SPAM_baseline	idle payload readout (Arm 8)	0.0000	[0.0000, 0.0005]
L_worst	max of the five raw leakages	**0.0112**	—
Δ_B	S_match − L_worst	**0.9696**	—
L_dest_corrected	L_dest − SPAM	**0.0087**	—
L_amount_corrected	L_amount − SPAM	**0.0101**	—
L_optype_corrected	L_optype − SPAM	**0.0082**	—
L_append_corrected	L_append − SPAM	**0.0112**	—
L_replay_corrected	L_replay − SPAM	**0.0002**	—

In-situ SPAM job (separate): Q_A p01 = 0.0024, Q_P p01 = 0.0000 (both ≤ 0.02). SPAM drift vs Arm 8 idle = 0.0000 (≤ 0.005). Gate **passed**.

Decision against preregistered rule (Fields 20–22)

PASS condition	Requirement	Observed	Met?
Destination changed	L_dest_corrected ≤ 0.02	0.0087	✅
Amount changed	L_amount_corrected ≤ 0.02	0.0101	✅
Operation changed	L_optype_corrected ≤ 0.02	0.0082	✅
Extra action appended	L_append_corrected ≤ 0.02	0.0112	✅
Approval replayed	L_replay_corrected ≤ 0.02	0.0002	✅
Reverification recovery	S_reverified ≥ 0.90	0.9781	✅
Boundary discrimination	Δ_B ≥ 0.70	0.9696	✅
In-situ SPAM	≤ 0.02 on both qubits	0.0024 / 0.0000	✅
Calibration drift	≤ 0.005	0.0000	✅

→ **PASS.** Every class of post-approval alteration and the replayed approval fail closed; a re-verified mutation executes. See `plots/arm_results.png` and `plots/integrity_discrimination.png`.

Secondary hypotheses (Field 19, descriptive — non-gating)

- **H2a (alterations fail closed at the SPAM floor):** point estimates read **False** for the four physical alteration arms. L_dest/L_amount/L_optype/L_append sit at ≈0.8–1.1%, a small residual **above** the pure-idle SPAM floor (0.00%) — consistent with the extra mid-circuit `reset` + re-measurement operations these arms carry, not with payload leakage. L_replay (0.02%) is statistically at the floor. All five remain far below the 0.02 ceiling, so the verdict is unaffected; the arms fail closed but are not indistinguishable from a bare idle qubit.

- **H2b (reverification concordance):** S_reverified (0.9781) and S_match (0.9808) — 95% Wilson intervals **overlap** → fresh re-verification of a mutated action restores execution within the matched-reference interval. Consistent with H2b.
- **H2c (replay ≈ physical alteration):** L_replay (0.0002) 99% interval does **not** overlap the four alteration arms — the replayed approval actually leaks less than the physically altered arms (it carries fewer error-accruing operations). Both mechanisms fail closed; they are not numerically identical on this hardware.

Interpretation boundaries (Field 23)

This is a **metrological characterization of a tamper-evident decision-to-execution binding** on this qubit pair (Q_A=5, Q_P=6) on ibm_marrakesh at this calibration (2026-07-16). It is **NOT new physics** and **NOT a cryptographic integrity guarantee**. The “action signature” is abstracted to a single committed bit realized via a two-phase (approve → fresh re-verify) dynamic circuit; the alteration arms are representative encodings of alteration classes unified by the commitment mismatch they induce. “Leakage” means unauthorized payload activation (payload firing when the executed action does not match the approved action), not computational-basis leakage. A PASS demonstrates that, on this hardware and binding, post-approval alterations fail closed and a re-verified action executes; it does **not** establish security against an adversary, generalize to other bindings/qubits/backends, or assert MAC/signature strength. Findings do not generalize without replication.

Relationship to prior work (Field 24)

ARK-441 (ibm_kingston) established the SPAM-resolved verify-then-execute (VBE) authorization boundary; ARK-446 replicated it cross-device on ibm_marrakesh; ARK-442 characterized its degradation under verification-to-execution **delay** (all PASS). ARK-444 extends the boundary from whether an authorization is valid to whether the executed action is exactly the approved action, binding the payload to a fresh execution-time verification and testing five post-approval alteration classes plus a reverification-recovery arm. The DENY-arm ceiling (L_corrected ≤ 0.02), the reverification floor (S_reverified ≥ 0.90), and the discrimination floor (Δ_B ≥ 0.70) carry over from ARK-441/442 and are all met here.

ARK-444 Results — Remnant Fieldworks Inc. / Derek Hone — 2026-07-16. Verdict PASS. Metrological characterization of a tamper-evident decision-to-execution binding; not new physics, not a cryptographic integrity guarantee. No Rescue After Failure.